

FCFS

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    int n, head, i;
    int queue[20];
    int totalSeek = 0;
    printf("FCFS Disk Scheduling\n");
    printf("Enter number of requests: ");
    scanf("%d", &n);
    printf("Enter request queue:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &queue[i]);
    }
    printf("Enter initial head position: ");
    scanf("%d", &head);
    printf("\nSeek Sequence:\n");
    printf("%d ", head);
    for(i = 0; i < n; i++) {
        totalSeek += abs(queue[i] - head);
        head = queue[i];
        printf("-> %d ", head);
    }
    printf("\n\nTotal Seek Time = %d\n", totalSeek);
    return 0;
}
```

```
1WA24CS218
FCFS Disk Scheduling
Enter number of requests: 8
Enter request queue:
98 183 37 122 14 124 65 67
Enter initial head position: 53

Seek Sequence:
53 -> 98 -> 183 -> 37 -> 122 -> 14 -> 124 -> 65 -> 67

Total Seek Time = 640

...Program finished with exit code 0
Press ENTER to exit console. 
```

SCAN

```
#include <stdio.h>
#include <stdlib.h>
void sort(int arr[], int n) {
    int i, j, temp;
    for(i = 0; i < n - 1; i++) {
        for(j = 0; j < n - i - 1; j++) {
            if(arr[j] > arr[j + 1]) {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}
int main() {
    int queue[20], n, head, diskSize;
    int i, j;
    int totalSeek = 0;
    int tempHead;
    printf("SCAN Disk Scheduling\n");
    printf("Enter number of requests: ");
```

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scanf("%d", &n);
printf("Enter request queue:\n");
for(i = 0; i < n; i++) {
    scanf("%d", &queue[i]);
}
printf("Enter initial head position: ");
scanf("%d", &head);
printf("Enter disk size: ");
scanf("%d", &diskSize);
if(head < 0 || head >= diskSize) {
    printf("Invalid head position\n");
    return 0;
}
for(i = 0; i < n; i++) {
    if(queue[i] < 0 || queue[i] >= diskSize) {
        printf("Invalid disk request: %d\n", queue[i]);
        return 0;
    }
}
sort(queue, n);
tempHead = head;
printf("\nSeek Sequence:\n");
printf("%d", head);
for(i = 0; i < n; i++) {
    if(queue[i] >= head)
        break;
}
for(j = i; j < n; j++) {
    totalSeek += abs(queue[j] - tempHead);
    tempHead = queue[j];
    printf(" -> %d", tempHead);
}
if(tempHead != diskSize - 1) {
    totalSeek += abs((diskSize - 1) - tempHead);
    tempHead = diskSize - 1;
    printf(" -> %d", tempHead);
}
for(j = i - 1; j >= 0; j--) {
    totalSeek += abs(queue[j] - tempHead);
    tempHead = queue[j];
}

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        printf(" -> %d", tempHead);
    }
    printf("\n\nTotal Seek Time = %d\n", totalSeek);
    return 0;
}

```

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1WA24CS218
SCAN Disk Scheduling
Enter number of requests: 8
Enter request queue:
98 183 37 122 14 124 65 67
Enter initial head position: 53
Enter disk size: 200

Seek Sequence:
53 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 199 -> 37 -> 14

Total Seek Time = 331

...Program finished with exit code 0
Press ENTER to exit console.

```

SSTF

```

#include <stdio.h>
#include <stdlib.h>
int main() {
    int queue[20], visited[20] = {0};
    int n, head, i, j;
    int totalSeek = 0;
    int tempHead;
    printf("SSTF Disk Scheduling\n");
    printf("Enter number of requests: ");
    scanf("%d", &n);
    printf("Enter request queue:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &queue[i]);
    }
    printf("Enter initial head position: ");

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scanf("%d", &head);
tempHead = head;
printf("\nSeek Sequence:\n");
printf("%d", tempHead);
for(i = 0; i < n; i++) {
    int min = 9999;
    int index = -1;
    for(j = 0; j < n; j++) {
        if(visited[j] == 0) {
            int distance = abs(queue[j] - tempHead);
            if(distance < min) {
                min = distance;
                index = j;
            }
        }
    }
    visited[index] = 1;
    totalSeek += min;
    tempHead = queue[index];
    printf(" -> %d", tempHead);
}
printf("\n\nTotal Seek Time = %d\n", totalSeek);
return 0;
}

```

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1WA24CS218
SSTF Disk Scheduling
Enter number of requests: 8
Enter request queue:
98 183 37 122 14 124 65 67
Enter initial head position: 53

Seek Sequence:
53 -> 65 -> 67 -> 37 -> 14 -> 98 -> 122 -> 124 -> 183

Total Seek Time = 236

...Program finished with exit code 0
Press ENTER to exit console.

```

C-LOOK

```
#include <stdio.h>
#include <stdlib.h>
void sort(int arr[], int n) {
    int i, j, temp;
    for(i = 0; i < n - 1; i++) {
        for(j = 0; j < n - i - 1; j++) {
            if(arr[j] > arr[j + 1]) {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}
int main() {
    int queue[20], n, head;
    int i, j;
    int totalSeek = 0;
    int tempHead;
    int index = 0;
    printf("C-LOOK Disk Scheduling\n");
    printf("Enter number of requests: ");
    scanf("%d", &n);
    printf("Enter request queue:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &queue[i]);
    }
    printf("Enter initial head position: ");
    scanf("%d", &head);
    sort(queue, n);
    tempHead = head;
    printf("\nSeek Sequence:\n");
    printf("%d", tempHead);
    for(i = 0; i < n; i++) {
        if(queue[i] >= tempHead) {
            index = i;
            break;
        }
    }
```

```

    }
    for(j = index; j < n; j++) {
        totalSeek += abs(queue[j] - tempHead);
        tempHead = queue[j];
        printf(" -> %d", tempHead);
    }
    if(index != 0) {
        totalSeek += abs(tempHead - queue[0]);
        tempHead = queue[0];
        printf(" -> %d", tempHead);
    }
    for(j = 1; j < index; j++) {
        totalSeek += abs(queue[j] - tempHead);
        tempHead = queue[j];
        printf(" -> %d", tempHead);
    }
    printf("\n\nTotal Seek Time = %d\n", totalSeek);
    return 0;
}

```

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1WA24CS218
C-LOOK Disk Scheduling
Enter number of requests: 8
Enter request queue:
98 183 37 122 14 124 65 67
Enter initial head position: 53

Seek Sequence:
53 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 14 -> 37

Total Seek Time = 322

...Program finished with exit code 0
Press ENTER to exit console.

```